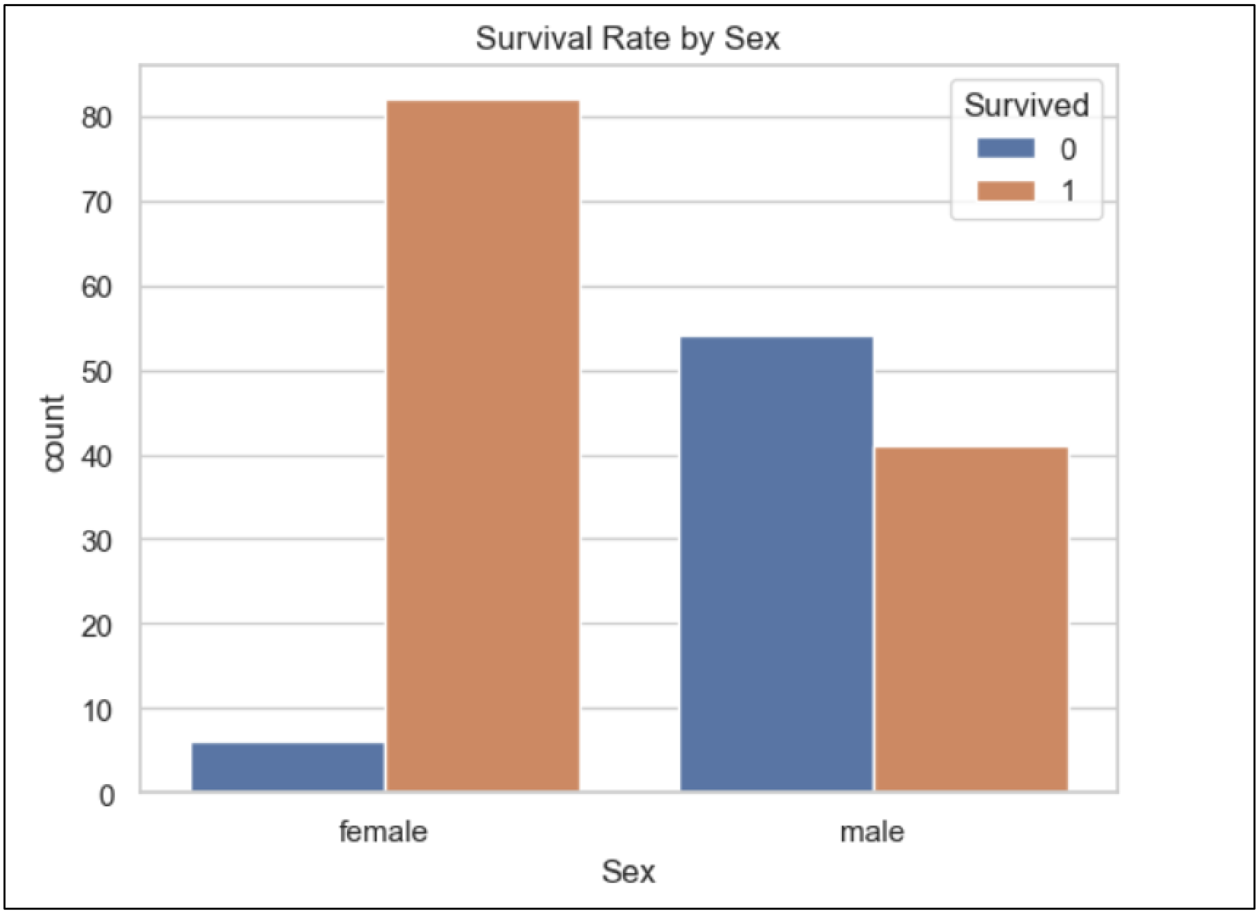
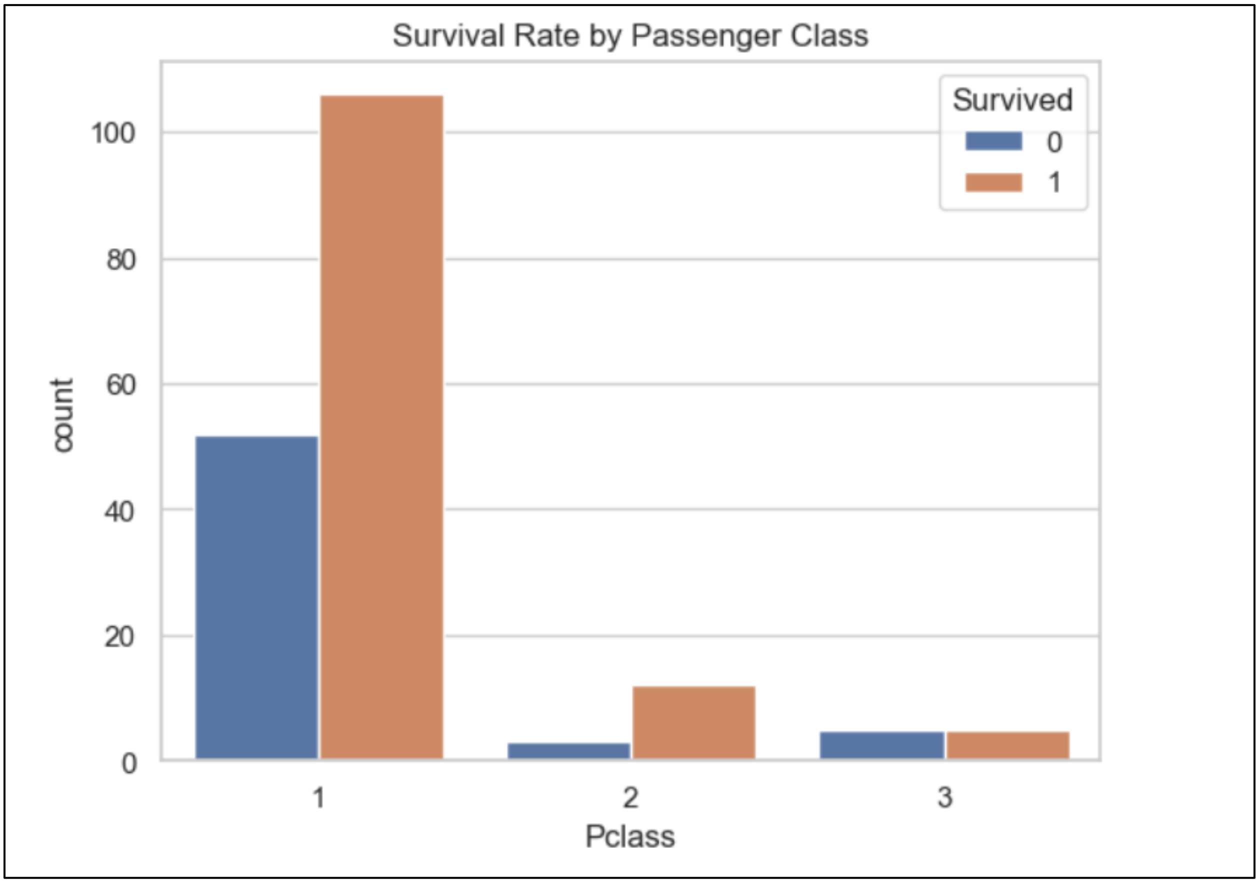
Task 5: Exploratory Data Analysis (EDA)

1. Survival by Gender



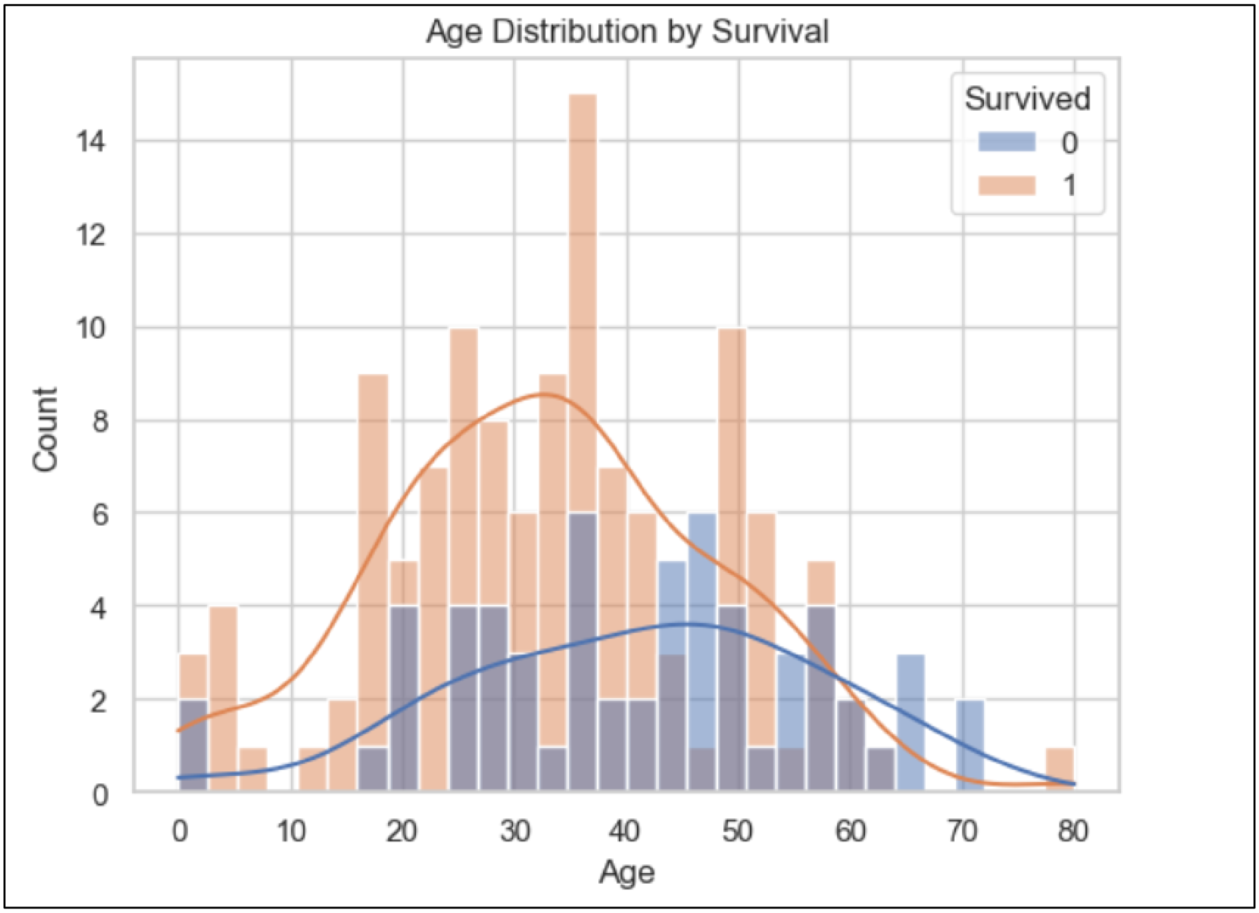
- **Observation**: A significantly **higher proportion of females survived** compared to males.
- **Conclusion**: Gender played a key role in survival rates — possibly due to the "women and children first" policy during evacuation.

2. Survival by Passenger Class



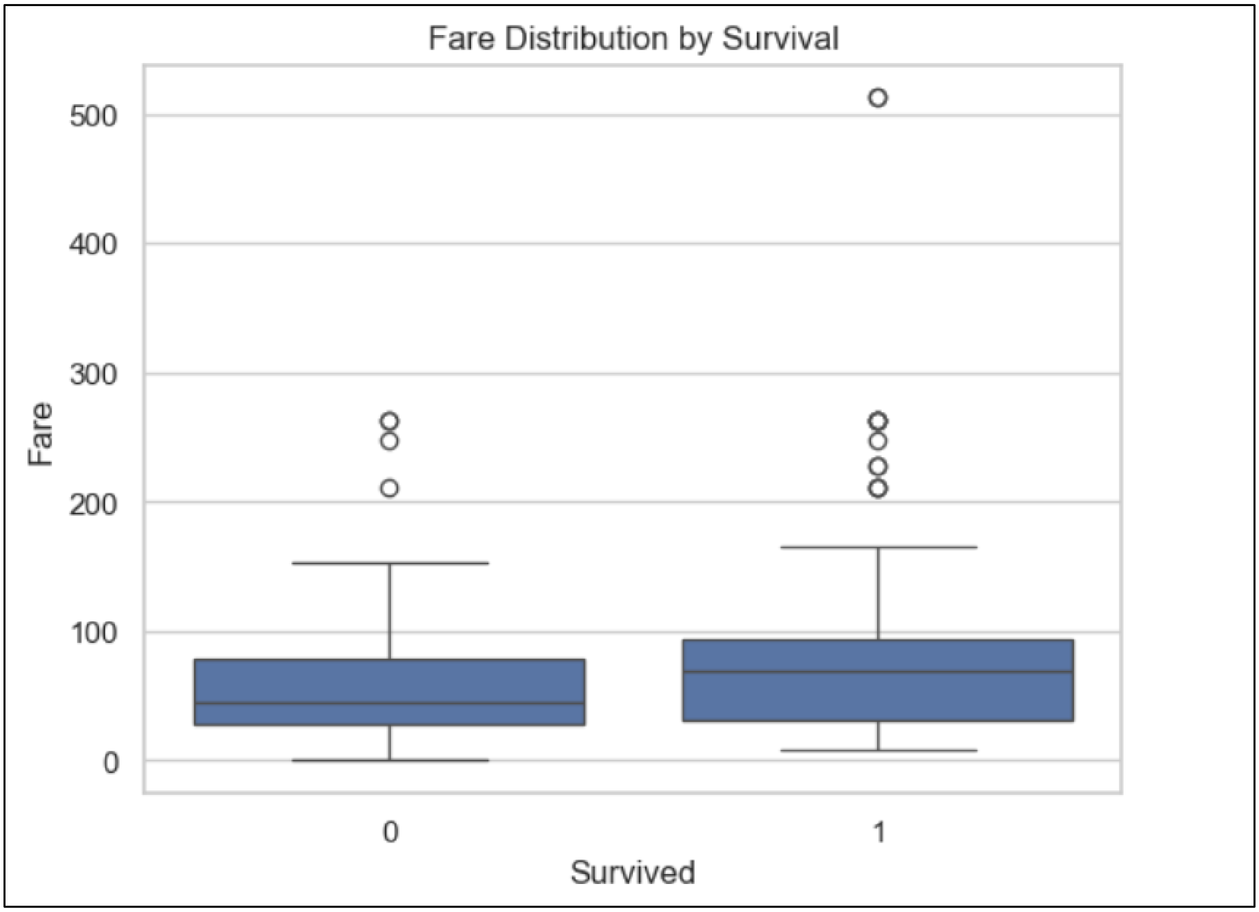
- **Observation**: Passengers in **1st class had the highest survival rate**, followed by 2nd class, with 3rd class having the lowest.
- **Conclusion**: **Passenger class correlated with survival**, indicating those with more resources or better access (e.g., cabins near lifeboats) had better chances.

3. Age Distribution and Survival



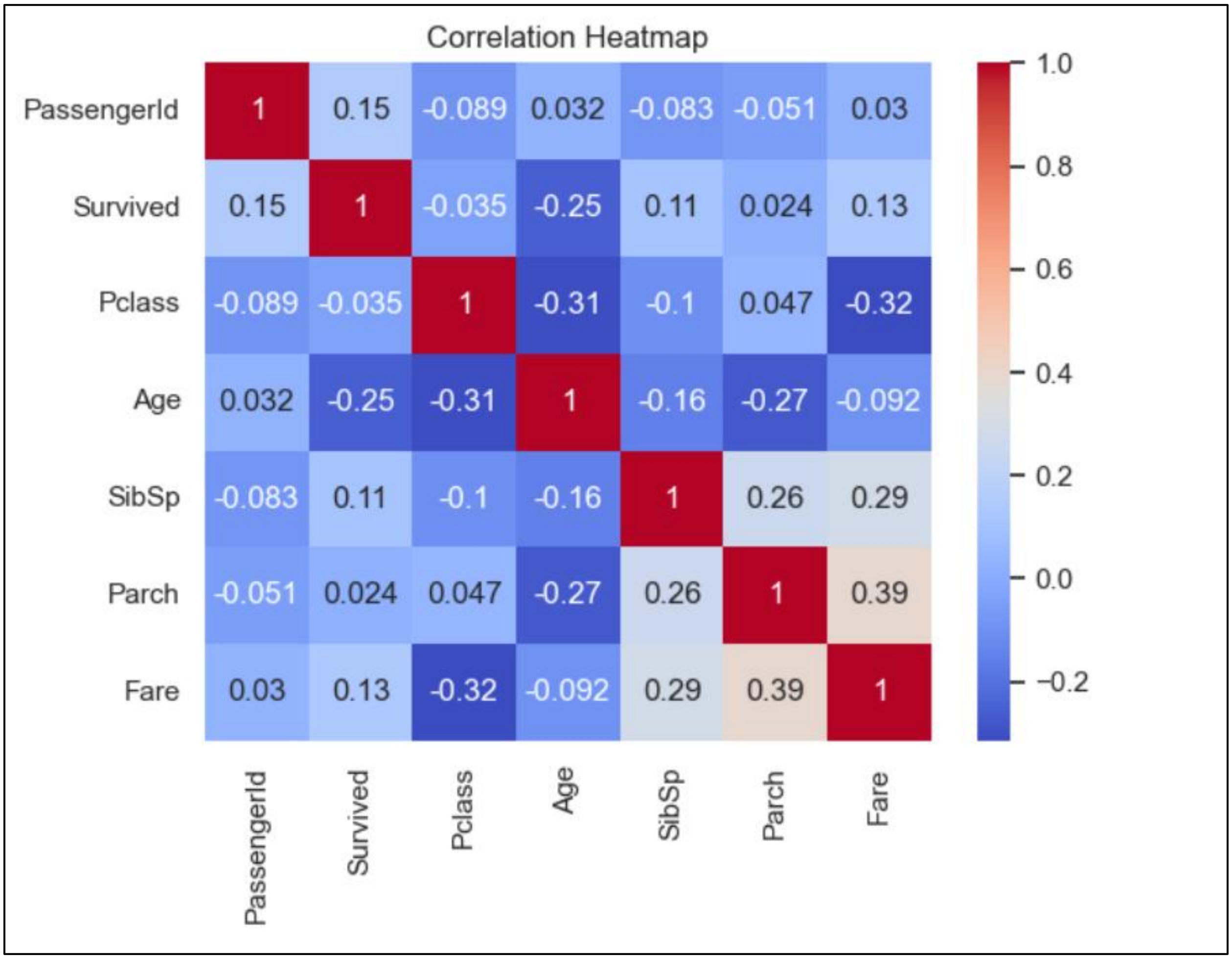
- **Observation**:
  - Children and young adults had **relatively higher survival**.
  - Elderly passengers had **lower survival rates**.
- **Conclusion**: There appears to be a **survival preference toward younger individuals**, especially children.

4. Fare Paid vs Survival



- **Observation**: The **box plot** shows survivors generally paid **higher fares**, with a broader range of fare values.
- **Conclusion**: **Higher fare passengers (likely 1st class)** had better access to safety, which aligns with the earlier class-based insights.

5. Correlation Heatmap Insights



- **Positive Correlations**:
  - Fare has a **positive correlation** with Survived (~0.26), supporting the idea that paying more meant better survival chances.
  - Pclass has a **negative correlation** with Survived (~-0.31) — higher class (numerically lower value) → better survival.
- **Age** and Survived correlation is weak, but visual evidence shows that **younger age groups** had better chances.